

Munem Shahriar Nijhum

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Location: Kushtia, Khulna, Bangladesh

PROFESSIONAL SUMMARY

Motivated Computer Science and Engineering student at Khulna University of Engineering & Technology (KUET) with a strong academic foundation built at Kushtia Zilla School and Notre Dame College, Dhaka. Passionate about software development, problem-solving, and building efficient, user-centred applications. Proficient in multiple programming languages including C, C++, JavaScript, and Python, with a keen interest in algorithms, data structures, and hardware implementation.

EDUCATION

Bachelor of Science in Engineering (BSc)

Khulna University of Engineering & Technology (KUET)

Department: Computer Science and Engineering

Expected Graduation: 2027

Higher Secondary Certificate (HSC) – 2022

Notre Dame College, Dhaka

Group: Science

Board: Dhaka

Result: GPA 5.00 (General Scholarship)

Secondary School Certificate (SSC) – 2020

Kushtia Zilla School, Kushtia

Group: Science

Board: Jashore

Result: GPA 5.00 (Talentpool Scholarship)

Junior School Certificate (JSC) – 2017

Kushtia Zilla School, Kushtia

Group: Science

Board: Jashore

Result: GPA 5.00 (Talentpool Scholarship)

Primary Education Completion (PEC) – 2014

Kushtia Zilla School, Kushtia

Result: GPA 5.00 (Talentpool Scholarship)

ACADEMIC ACHIEVEMENTS

- Stood **Second** in the **Primary Education Completion Examination 2014 (PEC)** in Kushtia Sadar Upazilla, Kushtia.
- Ranked **First** among four sections (Day-A & B, Morning-A & B) of **Kushtia Zilla School** from Class 6 to Class 10.
- Stood **First** in the **Junior School Certificate Examination 2017 (JSC)** in Kushtia District.
- Became **Divisional Champion** in the **Science Project (Junior Category)** hosted by Biggan Chinta, Prothom Alo at Khulna Public College (Class 9).

- Stood **Second** in Kushtia District and **Sixth (Combined Merit)** in the Jashore Board in the **Secondary School Certificate Examination 2020 (SSC)**.
- Ranked **87th (Top 100)** out of 2,000 students in the Science Group in the Sent-Up Examination at **Notre Dame College, Dhaka**.
- Received the **Dean's Award** (up to 2nd Year) for outstanding academic performance from the Department of Computer Science and Engineering, **KUET**.
- Received Class Representative (CR) Appreciation Certificate for serving as a CR of CSE-2k22 for two consecutive semesters 2-2 & 3-1 from the Department of Computer Science and Engineering, **KUET**.

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, JavaScript

Web Development: HTML, CSS, PHP, ASP.NET, Laravel

Tools & Technologies: Git, GitHub, Linux, VS Code, IntelliJ IDEA, Jira

Database: MySQL (basic), PL/SQL, Oracle

Other: Data Structures & Algorithms, Competitive Programming, Object-Oriented Programming, Advanced Programming, Web Programming

PROJECTS

5-Storey Lift Circuit Design

- Developed a Logisim-based counter circuit design project that accepts floor inputs from 0 (Ground Floor) to 4 (Top Floor).
- Implemented the counter circuit principle so that the 7-segment display shows intermediate floors when navigating between any two floors (e.g., 2nd to Top Floor displays 2, 3, 4; Top to Ground Floor displays 4, 3, 2, 1).

GitHub: github.com/nijhum682/Lift-circuit-designed-by-logisim

7-Segment Display Design

- Developed a Logisim-based 7-segment display circuit that shows corresponding decimal digits (0–9) based on binary input (000–111).

GitHub: github.com/nijhum682/Design-of-a-7-segment-display

FoodBike: Online Food Delivery System (Android Version)

- Built a Java-based food delivery application supporting menu selection, order placement, delivery issue reporting, returns, reviews, and search functionality.
- Integrated basic data validation to ensure input accuracy across all operations.

GitHub: github.com/nijhum682/FoodBikeAndroid

FoodBike: Online Food Delivery System (Desktop Version)

- Built a Java-based desktop food delivery system with the same core features as the Android version, including menu selection, order placement, and delivery management.
- Integrated basic data validation to ensure input accuracy across all operations.

GitHub: github.com/nijhum682/FoodBike

CV Builder

- Developed a Java-based CV Builder that collects user inputs (e.g., name, parents' names, phone number, address, photo, educational background, and technical skills) and generates a formatted Curriculum Vitae as output.

GitHub: github.com/nijhum682/Nijhum_2207111_CVBuilder

KUET CSE-22 Telegram Bot

- Developed a Python-based Telegram Bot that provides academic resources to CSE-22 batchmates, including class routines, slides, term and CT questions, syllabi, project files, batchmate information, and course details, accessible in any natural language.
- Extended the bot with off-topic Wikipedia queries, allowing users to ask about general topics freely.
- Deployed with 24/7 uptime via free AI hosting on the Vercel platform.

GitHub: github.com/nijhum682/CSE-22-Bot

Hardware Acceleration Club of KUET (HACK Portfolio Website)

- Developed an HTML and ASP.NET-based portfolio website for the departmental club HACK, featuring dedicated sections for Home, About, Activities, Achievements, Education, Projects, Technical Skills, and Profiles.
- Integrated themed background animations to enhance the visual experience.

GitHub: github.com/nijhum682/HACK-Portfolio

Self-Portfolio (Munem Shahriar Nijhum)

- Developed a personal portfolio website using HTML and CSS, with sections covering Home, About, Activities, Achievements, Education, Resources, Contact, and Profile.
- Integrated themed background animations to enhance the visual experience.

GitHub: github.com/nijhum682/my_portfolio

20-Bit Mini Computer (Using Logisim)

- Designed and simulated a 20-bit computer in Logisim using core components: PC, MAR, MBR, IR, AC, RAM, Fetch Flip-Flop, and a 20-bit ALU.
- Implemented a complete fetch–decode–execute cycle through a dedicated Control Unit.
- Developed a 20-bit ALU integrated with Booth's Multiplier, supporting ADD, AND, XOR, NOR, SHIFT, and COMPLEMENT operations.
- Defined custom instruction formats and opcodes for memory, branching, and processing instructions including ADD, LOAD, HLT, XOR, BUN, and NOR.

GitHub: github.com/nijhum682/20-bit-mini-computer

Numerical Calculator (Group Project)

- Developed a C++-based Numerical Calculator to solve common numerical problems using standard numerical methods and algorithms.
- Organised techniques such as root finding, equation solving, interpolation, and approximation into separate modules for better structure and maintainability.
- Designed with dedicated implementation, theory, input, and output files for each method, making it a practical educational tool.

Collaborators: Tahmid Hossain Chowdhury Mahin, Munem Shahriar Nijhum & Tamal Ghosh

GitHub: github.com/Tahmid2K22/Numerical-Calculator

Circuit Implementations (Hardware)

- 7-Segment Display Circuit: Constructed a physical 7-segment display circuit using ICs, jumper wires, LEDs, and a breadboard with power supply.
- RTL & DTL NAND & NOR Circuits: Built these circuits as part of a Digital Electronics project using ICs, jumper wires, LEDs, and a breadboard with power supply.
- Relay Circuit: Developed a relay circuit as part of an Embedded Systems project using Arduino Uno.
- Autonomous Car: Assembled a controlled autonomous car using required circuit components and Arduino Uno, programmed via the Arduino IDE.
- Obstacle Detector by Ultrasonic Sensor: Built a Obstacle Detector which detects obstacles by emitting a high-frequency sound wave and timing how long it takes for the echo to bounce back from the object, calculating the distance based on the speed of sound.

Wokwi Wi-Fi Controlled 7-Segment Display (First IoT Project)

- Built an IoT project that receives number inputs (0–9) from the Blynk mobile/desktop app and displays the corresponding digit on a 7-segment display via ESP32 Wi-Fi communication.
- Features include 7-segment display interfacing with ESP32, Wokwi simulation for virtual testing and debugging, and cloud-based IoT communication using Virtual Pin V0.

GitHub: github.com/nijhum682/WiFi-Controlled-7-Segment-Display

EduTrack - Online Learning Platform (Web & LMS Project)

- Built a feature-rich Learning Management System (LMS) designed to streamline education, enhance peer collaboration, and bring classroom management under one intuitive hub.
- Features include an interactive lecture player with likes/comments, a peer note exchange hub with teacher grading, a smart Q&A corner, live broadcasts, and an OpenAI-powered virtual tutor.

GitHub: <https://github.com/nijhum682/EduTrack.git>

CaseFlowX - Digital Police Case Management System (Web Application Project)

- Developed a digital case management system designed for police departments to streamline filing FIRs, managing records, and tracking investigations.
- Features include user authentication and role management, investigation tracking with a Scrum task board, evidence and criminal database management, a citizen's complaint portal, and an analytics dashboard.

Collaborators: Tahmid Hossain Chowdhury Mahin, Munem Shahriar Nijhum & Tamal Ghosh

GitHub: github.com/Tahmid2K22/CaseFlowX-Digital-Police-Case-Management-System

Travello - Online Bus Ticketing System (Database Project)

- Built a web-based bus ticketing application with an ASP.NET Core backend and Oracle Database integration to handle routing and booking management.
- Features include dynamic bus scheduling, route management, secure ticket reservation and seat booking, and database integration for transactional stability.

GitHub: github.com/nijhum682/Bus-Ticketing-System

IoT Based Automated Toll & Traffic Management System

- Developed a dual-purpose smart transportation system that automates toll collection and controls traffic flow using Wi-Fi-enabled microcontrollers and a web control panel.
- Features include automated toll deduction via ESP32, vehicle image capture using ESP32-CAM, real-time traffic signal optimization, and web dashboard integration for system monitoring.

Contributor: Munem Shahriar Nijhum & Tamal Ghosh

GitHub: github.com/Tamal-Ghosh/Automated-Toll-and-Traffic-Control-System-

ACTIVITIES

- Member – Special Group Interested in Programming Contest (SGIPC), KUET
- Member – Bit2Byte (Software Research & Development Community of KUET)
- Performance Analyst & Batch Representative – Hardware Acceleration Club of KUET (HACK)
- Volunteer – Bitfest (KUET CSE National Fest 2025)
- Research Member – Kushtia Science Research Club (KSRC)
- Host – 2nd KSRC Science Festival, Kushtia 2025
- Member – Notre Dame Science Club
- Member – Notre Dame Math Club
- Member – Kushtia Zilla School Science Club